

BI-W04

Lec 7, August 11

Conjugate prior: Normal–Normal (mean unknown, variance known)

$\theta \sim N(\mu_0, \tau_0^2)$. Should have high variance.

$$100 \text{ cm} - 200 \text{ cm}, \quad Y_1, \dots, Y_n \mid \theta \sim N(\theta, \sigma^2)$$

$$\pi(\theta \mid Y) \propto \frac{1}{\sqrt{2\pi\tau_0^2}} \exp\left\{-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right\} \cdot \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{-\frac{\sum(Y_i - \theta)^2}{2\sigma^2}\right\}$$

As $\tau_0 \rightarrow \infty$,

$$\pi(\theta \mid Y_1, \dots, Y_n) \approx \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{-\frac{(\theta - \bar{Y})^2}{2\sigma^2/n}\right\}$$

$$p(Y_1, \dots, Y_n) = p(Y_1, \dots, Y_n \mid \theta) \pi(\theta)$$

Examples of priors:

$$N(0, 100), \quad N(0, 10), \quad (Y_1, \dots, Y_n)$$

Regression illustrations:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

Does it need to be a valid probability density function?

Answer

Prior: **No**

Posterior: **Yes**

$$\pi(\theta \mid y) \propto \pi(\theta) \pi(y \mid \theta) \quad \pi(\theta \mid x) = \frac{\pi(\theta) \pi(x \mid \theta)}{\int \pi(\theta) \pi(x \mid \theta) d\theta}$$

$\pi(\theta \mid A) \propto \pi(\theta) \pi(A \mid \theta)$ need not integrate to 1.

But

$$\int \pi(\theta \mid x) d\theta = 1 \quad \text{if} \quad \int \pi(\theta) \pi(x \mid \theta) d\theta \text{ is finite } (< \infty).$$

Improper prior

$$\frac{\pi(\theta)}{\int_{-\infty}^{\infty} \pi(\theta) d\theta} \propto 1 \quad \Rightarrow \quad \int \pi(\theta) \pi(x \mid \theta) d\theta = \infty$$

$$\int \pi(\theta) \pi(x \mid \theta) d\theta < \infty$$

$$\int \pi(\theta) d\theta = C \leq \infty \quad \Rightarrow \quad \int \pi(\theta) \pi(x | \theta) d\theta \leq \infty$$

$$\frac{1}{C} \int \pi(\theta) \pi(x | \theta) d\theta \quad \int \pi(\theta, x) d\theta = 1$$

Example 1

$$X | \theta \sim N(\theta, 1), \quad \pi(\theta) \propto 1 \quad \Rightarrow \quad \int \pi(\theta) d\theta = \infty.$$

But after observing X ,

$$\pi(\theta | x) \propto \pi(\theta) f(x | \theta) \quad \Rightarrow \quad \pi(\theta | x) \propto \exp\left\{-\frac{(\theta - x)^2}{2}\right\}.$$

$$\theta | x \sim N(x, 1)$$

Example 2

$$X | \theta \sim U(0, \theta), \quad \pi(\theta) \propto 1.$$

$$\theta \sim N(\mu_0, \tau_0^2) \quad (\theta = \mu_0), \quad \frac{1}{\sqrt{2\pi\tau_0^2}} \exp\left\{-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right\}$$

$$\pi(\theta) \propto 1$$

$$\pi(x | \theta) = \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

$$\pi(\theta | x) \propto \pi(\theta) \pi(x | \theta) \propto \frac{1}{\theta} \mathbf{1}\{\theta > x\}.$$

$$\pi(\theta) \propto \frac{1}{\theta} \quad \longrightarrow \quad \text{proper posterior,}$$

$$\pi(\theta) \propto 1 \quad \longrightarrow \quad \text{improper posterior.}$$

$$\int \pi(\theta) \pi(x | \theta) d\theta < \infty.$$

$$\theta \sim N(\mu_0, \sigma_0^2).$$

$$\theta = \mu_0, \theta = \mu_0 + 10$$

$$\pi(\mu_0) = \frac{1}{\sigma_0 \sqrt{2\pi}}.$$

$$\pi(\mu_0 + 10) = \frac{1}{\sigma_0 \sqrt{2\pi}} \exp \left\{ -\frac{10^2}{2\sigma_0^2} \right\}.$$

$$\frac{\pi(\mu_0)}{\pi(\mu_0 + 10)} = \exp \left(\frac{10^2}{2\sigma_0^2} \right).$$

$$N(0, 10), \quad N(0, 100), \quad p(y_1, \dots, y_n)$$

$$\int \pi(\theta) d\theta \neq 1$$

$$\theta \sim N(0, 10) \quad \text{or} \quad \theta \sim N(0, 100)$$

Further

$$Y_1, \dots, Y_n \mid \sigma^2 \sim N(\mu, \sigma^2), \quad \mu \text{ known.}$$

What is $p(\sigma^2)$? (conjugate prior)

Binomial (Derive conjugate prior) $\rightarrow$ Express in terms of exponential family.

Poisson (Derive conjugate prior) $\rightarrow$ Express in terms of exponential family.

Derive $P(\lambda)$ using the representation of Poisson in exponential family of distributions.

$$\pi(\theta) \propto \exp \left\{ \eta(\theta)^T T(x) - A(\theta) \right\}$$

Hints

- Identify natural parameter, sufficient statistic.
- Construct prior over these natural parameters.
- Use Jacobian to convert that prior for the usual parameter of the distribution.
- (Hint: Think along the meaning of Beta prior for Bernoulli.)

Lecture 8, August 12, 2026

Poisson Regression with Covariate

Poisson regression with covariate.

$$(x_1, y_1) \quad (x_2, y_2) \quad \dots \quad (x_n, y_n)$$

$$(y_1, y_2) \quad \text{Priors \& Likelihood}$$

$$y_i \mid x_i \sim \text{Poisson}(\theta x_i)$$

$$p(y_1, \dots, y_n \mid x_1, \dots, x_n, \theta)$$

$$L(\theta) = \prod_i \frac{e^{-\theta x_i} (\theta x_i)^{y_i}}{y_i!}$$

$$L(\theta) \propto e^{-\theta \sum x_i} \theta^{\sum y_i}$$

$$L(\theta) \propto e^{-\theta \sum x_i} \theta^{\sum y_i}$$

- 1) Likelihood
- 2) Conjugate prior (or some other form)
- 3) Posterior
- 4) Predictive distribution

Non-informative Prior

$$\pi(\theta) \propto 1 \quad X \mid \theta \sim N(\theta, \sigma^2) \quad \pi(\theta) \propto 1 \quad \pi(y \mid \theta) \propto 1$$

$$\frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$$\frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} \quad \text{where } \theta = e^\gamma$$

Non-informative prior:

$$\phi = g(\theta) \quad \pi(\phi) = \pi(g^{-1}(\phi)) |J|$$

Jeffreys Prior

Jeffreys prior:

Any rule for determining the prior density should yield an equivalent result if applied to the transformed parameter, i.e. $p(\phi)$ computed by determining θ and applying (1) should match with the distribution that is obtained directly for $p(\phi)$.

Let $\phi = h(\theta)$ where h is a one-to-one transformation. Then,

$$p(\phi) = p(\theta) \left| \frac{d\theta}{d\phi} \right| = p(\theta) |h'(\theta)|^{-1}$$

Jeffreys principle says that $\pi(\theta) \propto \sqrt{I(\theta)}$ where $I(\theta)$ is the Fisher information for θ , i.e.

$$I(\theta) = -E \left(\frac{\partial^2 \log f(y | \theta)}{\partial \theta^2} \right)$$

$$\pi(\theta) \propto \sqrt{I(\theta)} = \sqrt{I(\phi)} \left| \frac{d\phi}{d\theta} \right|$$

Then,

$$I(\phi) = -E \left(\frac{\partial^2 \log f(y | \phi)}{\partial \phi^2} \right)$$

$$E \left[\left(\frac{\partial \log f(y | \theta)}{\partial \theta} \right)^2 \right] = I(\theta) \left| \frac{d\theta}{d\phi} \right|^2$$

$$\theta \rightarrow \theta + d\theta$$

$$\theta \in [\theta, \theta + d\theta] \rightarrow p(\theta) \approx 0$$

$$X \rightarrow p \quad p > 0.5 \quad \text{Var}(X) = \sqrt{\frac{N(1-p)}{p}}$$

$$p(\theta | X) \propto p(X | \theta + d\theta)$$

$$p(X | \phi) \propto p(X | \phi + d\phi)$$

Binomial

$$p = 0.5 \quad X | \theta \sim \text{Bin}(n, p) \quad \theta = \frac{p}{1-p} \quad \pi(\theta) \propto 1$$

Binomial

$$-E \left(\frac{\partial^2 \log f(y | \theta)}{\partial \theta^2} \right) = \frac{1}{p(1-p)}$$

Kullback–Leibler Divergence

K–L divergence (Kullback–Leibler divergence)

$$D_{KL}(P \parallel Q) = E_P \left[\log \frac{p}{q} \right]$$

$$E_\theta [d \log f(y \mid \theta) - \log f(X \mid \theta) d\theta]$$

$$\theta \rightarrow \theta + d\theta$$

$$d\theta^2 = I(\theta) d\theta^2$$

$$L(\theta) = \log f(y \mid \theta)$$

$$L(\theta + d\theta) = \log f(y \mid \theta + d\theta)$$

$$L(\theta + d\theta) = L(\theta) + L'(\theta) d\theta + \frac{1}{2} L''(\theta) d\theta^2 + o(d\theta^2)$$

$$-E_\theta [L'(\theta)] d\theta$$

$$-\frac{1}{2} E_\theta [L''(\theta)] d\theta^2 + o(d\theta^2)$$

$$D_{KL}(P(X \mid \theta), P(X \mid \theta + d\theta)) = 0 + I(\theta) (d\theta)^2 + o(d\theta^2)$$

$$\pi(\theta) \propto \sqrt{I(\theta)}$$

Poisson Example

$$\frac{e^{-\lambda} \lambda^x}{x!}$$

$$I(\lambda) = \frac{1}{\lambda} \quad \sqrt{I(\lambda)} = \frac{1}{\sqrt{\lambda}}$$

$$0 \rightarrow 2.25 \quad 2.25 \rightarrow 6.25$$

$$\sqrt{\lambda}$$